

Rohit Bandlamudi

Dallas, TX | 9728762159 | [LinkedIn](#) | bandlamudirohit@gmail.com

Summary

- Data Engineer with 3+ years of experience in **large-scale data migrations**, **pipeline optimization**, and **AI-driven automation** across cloud and on-premises systems.

Experience

Data Engineer, Copart

July 2025 to Present

- Scaled log data processing by migrating **~10B+ JSON-based event records** from **MariaDB to MongoDB**, handling **terabyte-scale data** and improving compound and time-range query performance by **5–12x**.
- Engineered **DB2 to MariaDB** migration using **Google BigQuery** as the transformation and staging layer via **Pentaho**, achieving validated source-to-target consistency across **70M+ records**.
- Reduced **query latency by 70 to 80%** on high-volume OLTP tables using indexing, partitioning, join optimization, and query restructuring.
- Implemented production **CDC replication pipelines** using **StarQuest** to stream data from MariaDB into BigQuery, enabling real-time dashboard delivery and downstream dimensional modeling on **GCP**.
- Reduced body shop data from **~12M to ~2.5M records** by orchestrating archival and deduplication pipelines using **Apache Airflow** to improve data quality.
- Designed **DataHub** metadata ingestion pipelines, custom transformers, and an **MCP-based discovery layer**, cataloging **4K+ database tables** and **2.5K+ Solr datasets** to enable **AI-assisted lineage**, governed metadata search, and faster self-serve discovery for engineering teams.
- Maintained data integrity across production systems by executing schema changes and data fixes using Git, Jenkins, and Liquibase with **controlled deployments** in an Agile based environment.
- Automated developer workflows by building **AI-driven tools** using LLMs and vector-based search for SQL generation, metadata discovery, and ticket triage.

Data and AI Engineer, Pyxeda

August 2024 - July 2025

- Designed **metadata ingestion workflows** aggregating data from source systems into **PostgreSQL**, transforming **~2M+ records** into structured JSON documents to power a **company-wide RAG system**.
- Built batch embedding pipelines using Sentence Transformers and PGVector, processing records in **80K-record batches** to handle API throughput constraints, enabling semantic search for over **200 users**.
- Orchestrated **weekly** ingestion and embedding refresh pipelines using Apache Airflow, implementing filtering and re-ranking layers to improve retrieval relevance.

Data Engineer, Intworks Solutions

July 2021 – July 2022

- Built **Python and SQL** based transformation workflows in **Azure Databricks**, processing datasets from **Azure Data Lake and Blob Storage** for ML training and inference pipelines.
- Improved data quality and consistency in **Azure Data Factory** pipelines by refining ingestion logic and dataset mappings, ensuring reliable inputs for downstream Databricks transformations.
- Engineered a **reusable Python preprocessing module** for encoding, normalization, and Gaussian transformations, standardizing feature preparation across **10+ ML use cases**.
- Implemented null checks, schema validation, and source-to-output reconciliation across transformation layers, reducing data inconsistencies by **30%**.

Skills and Certifications

- Languages: Python, SQL, Linux.
- Databases & Storage: MariaDB, MongoDB, PostgreSQL, MySQL, DB2, PGVector.
- Cloud & Data Platforms: Azure (Blob Storage, Data Factory, Databricks, Data Lake), BigQuery, DataHub.
- ETL and CI/CD: Apache Airflow, Pentaho, Liquibase, Git, Jenkins.
- AI and Visualization: LLMs, Sentence Transformers, Vector Search, RAG, LangChain, Tableau.
- Microsoft Certified Azure Data Scientist Associate November 2024 - November 2025
- AWS Certified Cloud Practitioner November 2023

Education

Master of Science in Business Analytics, The University of Texas at Dallas.

August 2022 - May 2024

- Relevant Coursework: Database Systems and Data Warehousing, Big Data Processing (Hadoop, MapReduce, PySpark, Kafka), Cloud Computing Fundamentals.